

Project Proposal: A Responsive Design Approach to Critical Media Literacy

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Rationale

This project addresses a critical gap in media literacy education for students in grades 6-9, who are increasingly exposed to digital misinformation, echo chambers, and manipulative online content during a critical stage of cognitive and social development. While it is targeted at students grades 6-9, our emphasis on usability will ensure that users outside the intended audience will find it useful. The approach will be grounded in the usability principles outlined by Issa and Isaías (2015), this project will emphasize all relevant aspects of usability, including: efficiency, effectiveness, and user satisfaction. Additionally, we will focus on educational usability, which encompasses alignment with learning goals, learner control, error recovery, and cognitive load management.

Woolgar's (1990) concept of 'configuring the user' will be employed to design critical reflections on how interface elements and interaction pathways influence users' behaviour and assumptions. The project, adopting a user-centered design approach, will assess usability through iterative trials with real users to uncover misalignments between design intentions and user experiences. As highlighted by Woolgar (1990), the trials will address the sociotechnical dynamics, primarily examining how tools can subtly position users into diverse roles, such as compliant learners, isolated explorers, or digital dependents.

The tool's overall design will be guided by principles of User Experience (UX) design, accessible design, and cognitivism through chunked microlearning modules and cognitive load management to create an inclusive learning tool for a diverse audience. By utilizing these principles, we aim to create a tool that accommodates the developmental attention capacities of early adolescents. The project thus aims to create an easy and responsive, pedagogically sound digital learning tool that is configured to support, rather than constrain its users.

Project Description

Our tool is a responsive Figma prototype designed to help students grades 6-9 critically engage with and digest digital and social media responsibly. Focused on nuanced topics such as: logical fallacies, misinformation, echo chambers, and rabbit-holes, the app is laser-focused on an efficient user journey with all extraneous elements stripped away. The app will not only act as a series of informative microlearning modules, but also as a quick reference tool that can help students navigate complex or even dangerous scenarios. UX design and usability are our primary focus, taking into heavy consideration the potential harm that poor implementation could inflict. Having identified a gap in youth media literacy, we believe a responsive app focused on helping young people navigate a world of rampant online dangers would truly be a net positive on society. The approach to achieving this will be two-pronged. The first being microlearning modules aimed to instruct on the topics we ultimately decide on with an estimated seat time of three to five minutes per module. Secondly, a series of “quick tools” each aimed at one topic with a requirement of an onboarded learner completing that tool’s narrowly-scoped purpose in under one minute. For example, the quick tool for misinformation might be a series of comparison screenshots of misinformation sectioned into categories containing key identifiers unique to each type and instructions on how to navigate each. An important consideration for usability is the effort to ensure the information we provide is apolitical and void of ideological zeal as we are not trying to change beliefs, rather provide the tools for students to be as informed as possible.

We chose Figma as our final delivery method as it allows us to build professional low-fidelity, aesthetic prototypes to showcase the functionality of our app on various screen types. Our scope will include responsiveness between two versions: desktop and mobile. Additionally, Figma allows for easy screen exports that we can embed into slides as well as a visually engaging platform for live demonstrating during the presentation.

Usability

As Woolgar emphasizes that technologies and systems are designed not just to serve users, but to actively configure them, to define user identities and constrain future actions (1990, p. 60), this has been considered in the design of our tool. By seeking to configure users' thinking through critical reasoning skills and social media analysis, our tool seeks to confront faulty logic often found in digital social media. By guiding users through a structured set of topics, users act as media detectives, skeptical consumers, critical thinkers, and emotionally aware participants, in such a way that configures their own interpretive reasoning skills both now and for the future. As each topic introduces new ways to challenge the user's existing knowledge of social media interpretation, the user will reflect on their understanding at the end of each segment allowing the concepts and ways of thinking to be further internalized. Bringing this back to Woolgar (1990), the usability of our tool is not measured by whether the user completes the pathway and trajectory of the task in and of itself, but rather by how effectively the user adopts the critical thinking frameworks provided through interacting with the tool. By checking for understanding along the way and assessing whether the user is applying logical reasoning through identifying media manipulation and reflecting on their own knowledge of what this looks like, the tool's usability is then assessed by how effectively it configures the user's cognitive skills and interpretive approaches beyond the learning process.

Adopting Woolgar's metaphor of the "machine as text" (1990), our usability assessment will treat our tool as a text we have authored. As "insiders" in the development process, we recognize that we inherently make assumptions about our "outsider" student audience, creating guardrails that configure the user by defining and delimiting their possible actions. Consequently, our usability trials will focus on observing how users interpret this text, moving beyond an assessment of our intended pathway to analyze their unique and emergent interactions. While our design provides structure, it is also intended to give users the agency to

forge their own paths. This approach aims to transform the user from a passive recipient of our configuration into an active co-configurator of their knowledge. The usability of our tool will thus be measured by its success in fostering this dynamic, where users actively shape their own learning rather than simply complying with the pathways we have built.

Educational Usability

The design of our media literacy tool for ages 12-14 integrates usability and Human-Computer Interaction (HCI) principles. According to Issa and Isaías (2015), educational usability focuses on optimizing user engagement and comprehension through intentional pedagogical design (Issa, 2015). Our tool will include chunking, structured pathways, and interactive elements for teachers, students, and parents to engage with. Our tool will implement these strategies while also addressing equity, diversity, inclusion, decolonization, and anti-racist (EDIDA) frameworks by offering culturally responsive content, customizable accessibility features, and inclusive design to support diverse learners. The app will include accessibility features such as compatibility across both desktop and mobile devices to support a wide range of learners and classroom environments. Students will improve retention by learning to critically analyze social media, understand diverse perspectives, and identify and challenge misinformation in respectful and informed ways.

From an educational standpoint, our app goes beyond surface-level interaction. It includes social media and misinformation, student note-taking capabilities, and real-time collaboration tools like comprehension checkpoints and gamified quizzes that reward student engagement. These features support differentiated instruction and help teachers stay organized while going paperless which improves sustainability and streamlines classroom communication. Families will have the option to monitor their child's learning progress and facilitate discussions on logical fallacies and misinformation found in social media to enhance school-home

connections. Thus, the goal of this project is to empower students to question misinformation and engage in constructive dialogue at school, at home, and in their personal day-to-day lives.

Grounded in cognitivism, this app will use chunking to present content in engaging, manageable segments to suit the digital habits and attention span of today's young learners. Usability testing with students and educators will be integral, aligning with HCI's call for iterative, user-centered design (Hewett et al., 1992). Our tool will be informed by this holistic approach that aims to use a student-and-teacher-centered approach to accommodate the digital habits and developmental needs of young users.

In an era where youth are immersed in digital environments, this project provides a timely and integrated educational solution. It seeks to enhance media literacy while empowering students to critically analyze misinformation, collaborate authentically, and navigate digital spaces with confidence.

References

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